INDUCED SYMBIOSIS THROUGH CONVEXITY

Four Conditions for Structural Alignment

and Their Failure Modes

Carpenter Buday

Independent Researcher

Abstract

We identify four conditions under which a convex cost function applied to participant deviation from a system's natural equilibrium, whose revenue returns to fund restoration, produces an endogenous intermediary class structurally aligned with system health. The critical condition --- loop closure --- is the property that prior near-equivalent mechanisms consistently violated. We prove asymptotic stability via Lyapunov analysis, establish intermediary emergence via a free-entry model, and prove that above-competitive extraction cannot persist indefinitely under economic entry. We characterise a distinct failure mode ---

the captured loop --- in which all four conditions are satisfied but the equilibrium target has been captured, producing alignment with identical stability properties aimed at a surrogate equilibrium. We prove necessity: violation of any single condition is sufficient to prevent structural alignment. A governance corollary follows: the closed convex loop performs rate-setting and threshold management continuously without lag, narrowing legitimate governance authority to bounded emergency response and periodic recalibration. The framework is demonstrated across two externally independent domains. A continuum game foundation (Appendix B) establishes that the aggregate dynamics emerge from individual dominant strategy behaviour.

Keywords: mechanism design, structural alignment, convex cost functions, captured loop, induced symbiosis, intermediary emergence, Lyapunov stability JEL Classification: D47, D82, D86, L10, G18

1. Introduction

Most mechanisms for producing alignment between individual incentives and collective outcomes rely on ongoing administration: rate committees that set prices, enforcement bodies that punish deviation, incentive programmes that reward compliance. These mechanisms share a structural property: remove the administration and they stop. The alignment they produce is contingent on continued human presence.

Most economic governance is not a solution to systemic fragility. It is evidence of it. A system that requires ongoing administration to survive is a system whose mechanism was never bonded to its natural equilibrium. Every governance layer added to a failing system --- every capital requirement, stress test, deposit insurance scheme, or central bank backstop --- is a response to a mechanism consuming its own substrate. Governance delays the countdown. It does not eliminate it. SVB did not collapse because governance failed. SVB was always collapsing. Governance had been running alongside it, extending T_collapse, for decades before the end arrived. This paper proposes a theory of that distinction: what separates mechanisms that must be propped up from mechanisms that sustain themselves, and why routing surplus out of the system that generates it --- extraction --- is the consistent structural feature of the former. This paper identifies a class of mechanisms in which alignment is not contingent. A convex cost function applied to participant deviation from a system\'s natural equilibrium, whose revenue returns to fund equilibrium restoration, produces an aligned intermediary class as a mathematical consequence of cost geometry. The intermediary class emerges without recruitment, earns by maintaining system health, and loses its business

model when it stops. This is forced geometry, not incentive design. Incentive design creates a gap between reward and behaviour that a sophisticated actor can exploit. Forced geometry relocates that gap from behaviour-enforcement to parameter-selection and measurement --- to (α, τ_max, τ_base, k, Φ, Ψ, R(t) measurement, R* calibration) --- where under defection-dominance with multiple capable providers (Proposition 8) it is narrower, structurally harder to sustain, and subject to geometric selection pressure rather than institutional enforcement. The gap is not eliminated. It is moved to a location where the mechanism's own dynamics work against its exploitation. This relocation is what the paper claims, and the remaining attack surfaces --- k-degradation (§7.2), Goodhart effects on R(t) (P2 discussion in §2.1), R* contamination in Regime B (§5.2), and parameter-setter incentives (§7.2 α_min discussion) --- are identified as scope limits rather than concealed within the forced-geometry framing. Correct behaviour remains the only behaviour consistent with a sustainable margin within the mechanism's operative domain. The mechanism produces alignment through cost geometry rather than requiring knowledge of agent types. Standard incentive-compatible mechanisms require the designer to know or elicit agent types to construct correct incentives. The closed convex loop requires no type knowledge --- the geometry self-enforces regardless of the type distribution F. Within the feasible range the intermediary is still making an individually rational choice, but the geometry eliminates the type-elicitation requirement that standard mechanism design cannot avoid. That is the precise claim: not the abolition of incentive compatibility, but the elimination of the type-knowledge prerequisite.

The specific property that prior near-equivalent mechanisms consistently violated --- and consistently failed to name --- is loop closure: the routing of revenue back into the system to fund equilibrium restoration. Tobin taxes, Pigouvian taxes, AMM fee structures, and EIP-1559 each satisfy three of the four conditions identified here. Each fails at exactly the same point: revenue exits the domain rather than returning to fund restoration.

We make six formal contributions: (1) the four-condition specification of the closed convex loop, with proof of local asymptotic stability and stochastic confinement; (2) Proposition 3, establishing sufficient conditions for endogenous intermediary emergence; (3) Proposition 4 (Leaching Impossibility), proving that above-competitive extraction cannot persist indefinitely under economic entry; (4) the captured loop stability result, showing that a closed loop converging to a surrogate equilibrium is dynamically as stable as a correctly aimed one, with a formal interior optimum for rational capture and a capture threshold d† characterising when the captured loop becomes worse than the open-loop alternative; (5) Propositions 8 and 9, establishing defection-dominance under adversarial entry and category

persistence across instantiation failure, formalising what we designate Tier 0 --- the space within which Tier 1 mechanisms emerge and replace each other; (6) Proposition 10 (Timescale Commensurability), establishing that structural alignment requires effective restoration rate at the fastest extraction timescale rather than at nominal measurement cycles.

The governance corollary follows from the mechanism\'s information-timing properties: in any domain where the closed convex loop operates, it performs rate-setting and threshold management functions continuously and without lag. What remains of legitimate governance authority is bounded emergency response and periodic structural recalibration. On falsifiability: the theory is falsifiable at every failure point --- every system that violates Condition 4 without collapsing, every captured loop whose intermediary class does not oppose recalibration, every domain satisfying all four conditions where intermediary emergence fails to occur falsifies a specific claim. What cannot currently be tested is the large positive confirmation, for reasons the theory itself predicts and which are addressed in §7.2.

1.1 Relation to Prior Literature

Hurwicz\'s original formulation asks whether a system can be designed so that individually rational behaviour produces collectively desirable outcomes. Subsequent results --- incentive compatibility, individual rationality, budget balance --- impose static constraints on equilibrium properties. The present paper asks a different question: under what conditions does rational behaviour continuously reproduce the conditions for good outcomes, without ongoing intervention? This is a strictly harder problem and the four conditions are its answer.

Bergemann and Morris (2005) establish robust mechanism design --- mechanisms that produce good outcomes without knowledge of participant types. The present mechanism achieves alignment without type elicitation: the geometry selects for aligned types through the survival constraint rather than eliciting type reports. Levin (2003) and Baker, Gibbons, and Murphy (2002) develop structural versus contingent alignment in relational contracting; the present contribution grounds this distinction in a specific geometric mechanism rather than repeated game dynamics. Rochet and Tirole (2003) on two-sided markets provides the closest adjacent framework to the intermediary emergence model; the key difference is that the intermediary class here is endogenous to the cost geometry rather than architected by a platform designer. The mean field game literature on competitive equilibria in large populations (Lasry and Lions 2007) provides the formal machinery for the multi-agent extension identified as future work.

Ostrom (1990) documented commons institutions that survived across centuries and identified common structural properties: graduated sanctions, monitoring systems, and community boundaries. This paper provides a formal account of why those properties correspond to the four conditions identified here. Ostrom was not finding designed systems --- she was finding survivors of geometric selection. The communities whose institutional evolution happened to approximate the four conditions persisted; the thousands that did not collapsed and left no record. Ostrom was doing paleontology of the geometry: working backward from the survivors to identify the selection pressure that produced them. The present framework provides the forward model that Ostrom's empirical method could not: given the conditions, here is what emerges and why. The captured loop concept extends the regulatory capture literature (Stigler 1971, Peltzman 1976) by establishing capture not as an institutional observation but as a convergence result: a closed loop aimed at a surrogate equilibrium produces the same stability properties as a correctly aimed one, making capture an equilibrium phenomenon rather than a pathology. The identification criterion --- the intermediary class opposes recalibration of the target --- is empirically testable.

Three additional bodies of literature bear directly on this paper. The revenue recycling and double dividend literature in environmental economics (Goulder 1995, Bovenberg and de Mooij 1994) examines whether environmental tax revenue should be returned to the system it regulates rather than used for general revenue --- the closest prior treatment of what this paper formalises as Condition 4. That literature evaluates revenue recycling as a policy choice; the present paper establishes loop closure as a necessary condition for structural alignment rather than a policy option, and provides the formal stability consequences of violating it. The dynamic mechanism design literature (Bergemann and Välimäki 2010, Pavan, Segal, and Toikka 2014) extends static mechanism design to settings where types and allocations evolve over time. The present paper is distinct in focus: rather than designing mechanisms for dynamic type revelation, it identifies the structural condition under which a cost function produces endogenous alignment without type elicitation at any point in time. The implementation theory literature (Maskin 1999, Moore and Repullo 1988) establishes conditions under which a social choice function can be implemented in equilibrium. The present contribution is complementary: it identifies a class of cost geometries that produce convergence to a specific equilibrium without requiring the designer to specify the social choice function or verify incentive constraints, provided the four conditions are satisfied. For readers in the game-theoretic tradition, the paper's results have the following correspondences. Proposition 3 is an equilibrium existence result for a competitive market with free entry, proved via Lyapunov methods rather than a fixed-point argument; both proof techniques establish the same object --- a stable equilibrium population n* --- with the

Lyapunov approach additionally characterising the convergence rate. Proposition 4 is the no-profitable-deviation condition for the intermediary market: above-competitive extraction triggers entry dynamics that drive profit to zero, making sustained extraction a dominated strategy under free entry. The paper's equilibrium concept is evolutionary stability (Maynard Smith 1982): the competitive equilibrium is the unique evolutionarily stable strategy of the intermediary game, resistant to invasion by both cheating strategies (which generate the advertising budget that recruits their own replacement) and over-optimisation strategies (which eliminate the spread that funds the niche). The cost geometry produces convergence results that are structurally analogous to dominant strategy equilibrium in the continuum formulation (Appendix B), but the primary equilibrium concept is evolutionary stability rather than Nash or Bayesian implementation. That formalisation, with heterogeneous agent types and strategic interaction, is identified as the primary direction for future work and would constitute a separate formal contribution.

2. The Mechanism: Conditions and Definitions

2.1 Four Necessary Conditions

Stated in domain-neutral terms first: a self-sustaining mechanism requires an attractor state defined by its own dynamics, discretion over agent behaviour that allows deviation from that attractor, a superlinear penalty on deviation that increases the cost of moving further away, and a pathway returning the extracted penalty back to the restoring force. The economic specialization follows.

Condition 1 --- Natural Equilibrium. The system has a stable resting state S* defined by its own internal dynamics. Formally: a fixed point of the system\'s autonomous dynamics f(x) such that ⟨f(x), −x⟩ \> 0 for x ≠ 0 in some neighbourhood --- the natural dynamics point toward S* from nearby states.

On the nature of S*. The closed convex loop does not create the structural equilibrium. It bonds to one that already exists. S* is ontologically prior to any mechanism: the rate at which surplus and deficit clear in credit markets, the load distribution at which a grid is stable, the population size at which a fishery regenerates at the rate it is harvested. These attractors exist whether or not anyone builds a mechanism around them. The mechanism does not invent them. It stops fighting them and attaches human economic activity to them geometrically. This is why the Self-Location Observation (§3.1) has the character it does: S* does not need to

be specified because it is already there. The mechanism finds it the way water finds a drain. The designer's task is not to create the attractor but to build a cost geometry that bonds participant activity to it without distorting it. Ostrom's surviving commons were not communities that designed correct equilibria. They were communities that stopped fighting the ones that existed and organised their institutions around them. The ones that invented their own targets collapsed. This ontological priority of S* motivates a clarification in the tier taxonomy: S* is the pre-existing attractor --- the domain's natural equilibrium, present before any mechanism is designed. The closed convex loop is the bonding mechanism that attaches human economic activity to S* without distorting it. Grid frequency at 60 Hz is the attractor; the convex deviation charge with revenue routing to stabilisation is the bonding mechanism. The sustainable yield rate of a fishery is the attractor; a convex harvesting cost with revenue returning to stock restoration would be the bonding mechanism. The mechanism does not create the attractor. It stops fighting it.

Three domain prerequisites must hold for the full induced symbiosis result. P4 --- Substrate Reversibility: the domain's restoration function Ψ can reduce \|x\| without permanent physical depletion. Formally: the restoration cost per unit of \|x\| reduction is bounded and does not grow without limit as cumulative system activity increases. P4 is the physical filter beneath the four conditions. In abstract domains --- monetary systems, credit markets, DeFi, information markets --- restoration is rebalancing: moving numbers, adjusting obligations, reallocating existing claims. The restoration cost per unit is constant. P4 holds trivially and Tier 1 is achievable. In physical domains with regenerative cycles --- forestry, fisheries --- restoration cost per unit is bounded but depends on the biological regeneration timescale. P4 holds approximately within that timescale; Tier 1 is achievable provided the operating horizon is commensurate with the regeneration cycle. In extractive physical domains --- mineral extraction, fossil fuels --- restoration requires reversing entropy at geological timescales. Formally: let E(t) denote cumulative extraction and C_r(E) the marginal restoration cost per unit of \|x\| reduction. In extractive physical domains, lim_{E → E_max} C_r(E) = ∞, where E_max is the finite extractable reserve. The restoration cost is unbounded as the substrate approaches depletion. P4 fails. The loop cannot close. The complement definition's requirement that a domain be "healed by the penalty that behaviour incurs" implicitly requires P4: healing must be physically possible. Where P4 fails, the cost function can still impose genuine systemic cost (Condition 3) but the restoration operator Ψ has no substrate to work with. Tier 1 is categorically unreachable. Tier 2 remains achievable as a managed approximation, with T_collapse bounded by the substrate's depletion rate rather than by the mechanism's governance quality.

A note on Condition 1 across domains. Two distinct regimes apply. Regime A covers domains with physically or structurally defined equilibria --- electrical grid frequency at 60 Hz, reserve ratios with physics-determined solvency thresholds --- where S* is objectively identifiable and Condition 1 is cleanly satisfied. The mechanism amplifies a pre-existing structural attractor. Regime B covers domains where the target is historically calibrated or institutionally defined --- interbank leverage thresholds, DeFi reserve ratios --- where the mechanism's principal contributions are the alignment and emergence machinery: whatever target is calibrated, the geometry produces structural symbiosis with it. The R* contamination risk is specific to Regime B. This distinction is maintained throughout the domain applications in §5.

Condition 2 --- Participant Discretion. Participants exercise genuine discretion over the intensity, timing, or volume of their system interactions. This creates the variance in behaviour that the convex cost function can price differentially.

Condition 3 --- Genuine Systemic Cost. Participant deviation from equilibrium imposes genuine cost on the system --- cost that the mechanism can measure and price through the cost function. The formal minimum requirement is unidirectional: participant deviation harms the system. Bidirectional coupling (system degradation also harms participants) strengthens resilience but is not formally required by the proofs.

Condition 4 --- Loop Closure. Revenue generated by the convex cost function returns to fund equilibrium restoration rather than exiting the domain. This is the condition that prior near-equivalent mechanisms consistently violated. Condition 4 is not merely a technical requirement for stability. It is the metabolic condition: the difference between a mechanism that feeds itself and one that feeds on itself. A mechanism satisfying C4 routes surplus back to its own substrate --- it is self-nourishing. One violating C4 routes surplus out --- it is self-consuming. The former can survive indefinitely. The latter carries a countdown from the day it was built, hidden inside the governance and management that slow the drain.

Three domain prerequisites must also hold for the full induced symbiosis result: (P1) timescale commensurability --- the cost signal and the equilibrium it references operate on commensurable timescales; (P2) state variable observability --- the deviation metric R(t) is observable and manipulation-resistant; (P3) market scale --- $S \cdot V > F$, the spread multiplied by addressable volume exceeds the fixed cost of intermediary infrastructure.

A clarification on P1: most interesting domains have multiple timescales operating simultaneously --- a grid has dynamics on millisecond, minute, day, and year scales; a financial system has dynamics on transaction, quarterly, and generational scales; the

biosphere has dynamics on metabolic, seasonal, and geological scales. A closed convex loop calibrated to one timescale may be blind to deviations on another. This is a distinct failure mode from capture as currently defined: capture can occur by exploiting timescale mismatches --- extracting at a fast timescale while the mechanism's measurement and response operate at a slower one. P1 should therefore be understood not as a binary condition but as a requirement that the mechanism's response timescale be commensurable with the fastest extraction timescale in the domain. Violation of this commensurability requirement produces a timescale-mismatch capture that may not trigger the identification criterion of Definition 7 if the intermediary class operates faster than the measurement apparatus.

A clarification on P2: the requirement that R(t) be observable and manipulation-resistant does not distinguish between cases where R(t) is cheaply observable (grid frequency at every point on the network, container return rate from deposit records) and cases where R(t) is expensive to measure, contested, or manipulable (biospheric deviation from equilibrium, systemic financial risk, social cohesion). When the cost of measuring R(t) is significant relative to the cost of the mechanism itself, a distinct class of failure modes opens: the mechanism converges to whatever the measurement apparatus reports rather than to the true system state. This is Goodhart's law applied to the closed convex loop --- the deviation metric ceases to be a measure of the system state and becomes a target in its own right, and the mechanism's convergence guarantee becomes a guarantee of convergence to the metric rather than to S*. The deposit-refund and grid cases work partly because the deviation metric is trivially measurable. Most domains where one would want to apply the framework --- climate, biodiversity, financial stability --- have R(t) that is expensive, contested, and manipulable. The framework's predictions in these domains are conditional on a measurement infrastructure the framework does not specify and whose quality determines whether the mechanism converges to S* or to a Goodhart surrogate.

2.2 The Closed Convex Loop --- Formal Definition

Definition 1 (Natural Equilibrium). A state S* such that in the absence of external perturbation, the system\'s autonomous dynamics return it toward S* from nearby states, with restoring forces that strengthen with distance.

Definition 2 (Convex Cost Function). C: $\mathbb{R} \to \mathbb{R}_+$ is convex with C(0) = τ_base ≥ 0, C\'(0) ≥ 0, and C\'\'(x) \> 0 for x ≠ 0. In practice: C(x) = min(τ_max, τ_base + c\|x\|\^α) for α \> 1, applied to x = R(t) − R*.

Definition 3 (Closed Convex Loop). A tuple (S^*, C, R, Φ) where S^* is a natural equilibrium, C is a convex cost applied to deviation $x = R(t) - R^*$, and Φ is a revenue-routing map returning collected revenue to a restoration operator Ψ that reduces $\|x\|$. The loop is closed if revenue does not permanently exit the domain.

Definition 4 (Intermediary Class). A population of agents who earn by exploiting the cost differential between optimised and unoptimised participant behaviour, with business models structurally dependent on system health.

Definition 5 (Structural Alignment). The property that an agent's profit motive is identical to the system's interest by construction --- not because the agent values the system's health but because the geometry makes correct behaviour the only behaviour consistent with a sustainable margin. The permanence requirement --- permanently, not contingently --- is independently motivated by the paper's central question: does the mechanism still work if everyone administering it disappears? A mechanism whose alignment holds only while external conditions cooperate is a managed system. The permanence requirement is the formal line between the two. Condition 4's necessity follows from this independently: without funded restoration, the sustainable margin condition holds only while $\sigma < \sigma_{crit}$, which depends on factors outside the mechanism's design. The necessity proof shows C4 is what makes the independently-motivated permanence property hold --- it is not circular.

Definition 6 (System Dynamics). $dx/dt = F(x) - k\cdot\Phi(C(\|x\|)) + \varepsilon(t)$, where $F(x)$ are the system's natural participant forces satisfying $\langle F(x), -x\rangle > 0$ by Definition 1 ($F(x)$ is a restoring force, co-directional with the mechanism term $-k\cdot\Phi(C(\|x\|))$), $k > 0$ is the restoration efficiency coefficient of the mechanism, $k_F > 0$ is the natural restoring coefficient of $F(x)$ alone defined by $\|F(x)\| \geq k_F\cdot\|x\|$ for $\|x\|$ in a neighbourhood of 0, and $\varepsilon(t)$ is stochastic disturbance.

Definition 7 (Captured Loop). A closed convex loop in which all four conditions are satisfied but the equilibrium target S^* has been replaced by a surrogate equilibrium $S_c \neq S^*$ serving the interests of a dominant intermediary or captured governance authority. The captured loop is dynamically as stable as a correctly aimed loop. Identification criterion: the intermediary class opposes recalibration of the target. Note: the underlying dynamics --- surrogate equilibrium stability, intermediary opposition to recalibration --- can appear in systems that only partially satisfy the four conditions; the formal definition requires all four.

Proposition 1 (Induced Symbiosis). A closed convex loop (Definition 3) satisfying Conditions 1--4 produces as an emergent property an intermediary class (Definition 4) whose profit motive is structurally aligned (Definition 5) with equilibrium maintenance. This

alignment is not designed. It is a mathematical consequence of cost function geometry. \[Established jointly by Propositions 3--6: Proposition 3 establishes emergence, Proposition 4 establishes durability under free entry, Proposition 5 establishes that aggregate dynamics arise from individual dominant strategy play, and Proposition 6 establishes necessity. The formal arguments follow in §3.\]

Remark 1 (Three-Tier Classification)

Systems can be classified by the depth at which their boundary conditions are self-enforcing:

Tier 1 --- Full mechanisms. Boundary conditions are self-enforcing by structure. Entry is open because the geometry itself makes exclusion physically or mathematically impossible. No intervention required at any layer.

Tier 2 --- Mechanisms in managed shells. The inner loop is a true mechanism. The boundary conditions require legal or institutional enforcement to remain open. The mechanism runs without intervention given the shell. The shell requires management.

Tier 3 --- Managed systems throughout. No self-enforcing layer exists anywhere. Intervention is required at every level. The mechanism design tradition has almost entirely produced Tier 3 systems.

A true mechanism is only as deep as its shallowest managed dependency. This paper identifies a class of mechanisms that does not require ongoing intervention to sustain alignment, and characterises the conditions under which that class is non-empty. Most broken economic systems are Tier 3. Moving them to Tier 2 does not require eliminating governance. It requires installing a closed convex loop inside the governance structure that already exists. Tier 2 is not an acceptable steady state. It is a transitional structure --- the correct first move in a world where Tier 3 institutions control the regulatory environment, enabling the inner loop to demonstrate its properties while the managed shell provides political cover. The objective is Tier 1: boundary conditions self-enforcing by structure, no managed dependencies remaining.

2.3 Tier 0: The Space of Instantiations

The three-tier classification of Remark 1 distinguishes individual mechanism instantiations by the depth at which their boundary conditions are self-enforcing. The taxonomy as stated is complete with respect to instances --- every instantiated mechanism falls into one of the three tiers --- but it does not capture a distinct structural category the framework implies: the

space within which Tier 1 instantiations emerge, compete, and replace each other when individual instances fail. We designate this category Tier 0.

Definition 9 (Tier 0 System). A Tier 0 system is a domain satisfying Conditions 1--4 at the category level such that the set of possible Tier 1 instantiations within it is non-empty and self-replenishing under Proposition 3. Formally: a domain D with aggregate natural equilibrium S*_D, participant discretion over which specific instantiation to use, genuine systemic cost for the absence of any instantiation, and loop closure at the category level --- where loop closure means that the geometric niche the category occupies does not permanently collapse when individual instantiations fail.

The distinction between Tier 0 and Tier 1 is not scale. A Tier 0 system is not a large Tier 1 system. Tier 1 through Tier 3 classify instantiated mechanisms by internal self-enforcement. Tier 0 classifies the domain in which instantiations of a given type can exist, compete, and replace each other. Tier 1 stability is the asymptotic stability of Proposition 1: convergence of a single instance to its equilibrium S*. Tier 0 stability is the population-level result that follows from applying Proposition 3 and Proposition 8 recursively at the instance level rather than the agent level. When an instance fails, the unmet demand for the Tier 0 function produces a spread that recruits replacement instances by the same emergence dynamics that Proposition 3 establishes for intermediaries within a single mechanism.

Proposition 9 (Category Persistence). A Tier 0 domain satisfying Definition 9 with capability-provider set of cardinality greater than one exhibits persistence of the category under arbitrary failure or suppression of individual instantiations. Proof. Let M_fail be any subset of instantiations that fail at time t. The residual demand creates a spread observable to potential entrants. By Proposition 3 applied at the mechanism level, entry occurs whenever S_D times residual volume exceeds F_D. By Proposition 8 applied at the mechanism level, no actor can durably prevent entry provided the capability-provider set contains more than one member. Therefore the category persists: failure of any subset of instantiations triggers replacement dynamics that restore aggregate capacity.

Tier 0 domains can be nested. A Tier 0 category can itself be an instantiation within a larger Tier 0 domain, producing a recursive structure in which persistence claims apply at each level of nesting independently. This recursive structure clarifies the role of P4 (substrate reversibility). Under the Tier 0 formulation, P4 is more precisely understood as a constraint that the enclosing Tier 0 system imposes on the Tier 1 instantiations within it. A Tier 0 domain whose aggregate substrate approaches depletion cannot sustain its constituent Tier 1 instantiations regardless of how well each instance satisfies Conditions 1--4 internally.

The Tier 0 vocabulary formalises what designer selection (§7.4) operates on. Designer selection is the Proposition 3 dynamics applied at the mechanism-design level within a Tier 0 domain. Designers who specify the correct S* produce Tier 1 instantiations that survive; designers who specify surrogate equilibria produce captured loops that persist until d-dagger is crossed and then collapse. Over sufficient iterations, the surviving class is the one whose geometric properties most closely approximate the four conditions. This is the forward model that §1.1 attributes to the framework --- stated now in explicit evolutionary terms.

3. Formal Results

3.1 Stability of the Deviation Dynamics

Self-Location (Observation). The closed convex loop is self-locating with respect to S*: given correct specification of R*, the mechanism converges to S* through its own operation, since $x = R(t) - R^*$ and the restoration operator drives $x \to 0$. Incorrect specification of R* produces the captured loop (Definition 7). This follows directly from the definitions and is stated here to name the failure mode rather than as a formal proposition.

3.1a Deterministic Lyapunov Proof

Let $x(t) = R(t) - R^*$ denote scalar deviation from equilibrium. For a convex cost $C(x) = c\|x\|^\alpha$ with $\alpha > 1$, the net restoration dynamics are:

$$dx/dt = -k \cdot c \cdot \|x\|^\alpha \cdot \mathrm{sign}(x)$$

Lyapunov candidate $V(x) = \|x\|^{(\alpha+1)}/(\alpha+1)$. Then $V(0) = 0$, $V(x) > 0$ for $x \neq 0$, and:

$$\dot{V}(x) = -k \cdot c \cdot \|x\|^{(2\alpha)} < 0 \text{ for all } x \neq 0$$

The operating domain D is defined as $D = \{x : \tau_{base} + c\|x\|^\alpha \leq \tau_{max}\} \cap \{x : V(x) \geq V_{min}\}$. The first constraint ensures the cost function remains strictly convex and provides positive restoring force --- for $\|x\|$ beyond the τ_{max} threshold the cost cap binds and the restoring force from the mechanism is zero. The second constraint ensures the restoration budget $c\|x\|^\alpha \cdot V(x) \geq c\|x\|^\alpha \cdot V_{min} > 0$. Both constraints are satisfied throughout D by construction. The equilibrium $x = 0$ is asymptotically stable within D. To verify this holds under the full dynamics including F(x), apply the same Lyapunov candidate $V(x) = \|x\|^{(\alpha+1)}/(\alpha+1)$ to the complete system $dx/dt = F(x) - k \cdot c \cdot \|x\|^\alpha \cdot \mathrm{sign}(x)$. Computing

$\dot{V}$ under the full dynamics: $\dot{V}(x) = |x|^{\alpha}\cdot\mathrm{sign}(x)\cdot[F(x) - k\cdot c\cdot|x|^{\alpha}\cdot\mathrm{sign}(x)] = -k\cdot c\cdot|x|^{(2\alpha)} + |x|^{\alpha}\cdot\mathrm{sign}(x)\cdot F(x)$. By Definition 1, $\langle F(x), -x\rangle > 0$ for all $x \neq 0$, which means $\mathrm{sign}(x)\cdot F(x) \leq 0$ --- the natural forces point toward $x = 0$. Therefore the second term is non-positive and $\dot{V}(x) \leq -k\cdot c\cdot|x|^{(2\alpha)} < 0$ for all $x \neq 0$. The full dynamics produce strictly faster convergence than the restoration term alone. The characteristic decay is polynomial at rate $t^{(-1/(\alpha-1))}$, not finite-time. The convergence is asymptotic rather than finite-time, which is the correct characterisation for a mechanism operating under continuous disturbance.

A clarification on Definition 1: the requirement $\langle F(x), -x\rangle > 0$ means the system's autonomous dynamics already tend toward S*. The mechanism does not create this tendency --- it amplifies it, and crucially it produces the intermediary class (Proposition 3) whose survival depends on the system remaining near S*. The mechanism's contribution is therefore alignment (Definition 5), not merely stability. A system satisfying Condition 1 but not Conditions 2--4 may converge to S* under sufficiently weak disturbances, but without the intermediary class no agent's survival is structurally identical to the system's health. Structural alignment requires both convergence and the endogenous intermediary class. Note: this stability result applies equally to the captured loop. Convergence toward S_c is asymptotically stable under identical conditions, since the Lyapunov argument requires only convexity about the target and loop closure --- not that the target is the structural equilibrium.

3.1b Stochastic Extension

For the SDE $dx_t = -k\cdot c\cdot|x_t|^{\alpha}\cdot\mathrm{sign}(x_t)\,dt + \sigma\cdot dW_t$, Itô's formula with $V(x) = ½x^2$ yields:

$$d/dt \cdot E[V(x_t)] = -k\cdot c \cdot E[|x_t|^{(\alpha+1)}] + ½\cdot\sigma^2$$

The practical stability neighbourhood satisfies $|x_{\infty}| \sim (\sigma^2/(2kc))^{(1/(\alpha+1))}$. Numerical validation across $\alpha \in \{1.5, 2.0, 3.0, 3.5\}$ and $\sigma \in [0.1, 2.0]$ confirms the bound throughout the specified parameter range, with safety margins 13.7%--30.7% (Appendix A).

3.1c Scope

The proofs establish local asymptotic stability and practical stability under continuous disturbance. Global stability within a single instantiation (Claim B) remains open. This is less load-bearing than it appears: the paper's persistence claims require only that when one instantiation fails, a new one emerges --- Claim C, established by Proposition 3 and the

counter-cyclical spread property. A mechanism can be locally fragile and globally persistent simultaneously.

3.2 Intermediary Emergence

Define the spread $S := E[C(q_{unopt})] - C(q_{opt})$, fixed cost of intermediary infrastructure F, and addressable volume V. The baseline model treats S as constant in n. In practice, as n rises, competitive pressure on intermediary margins compresses S: $S(n)$ is weakly decreasing in n. Under spread compression, the equilibrium population $n^* = S(n^*)\cdot V/F$ is smaller than the baseline predicts but remains positive provided $S(n) > F/V$ for some $n > 0$, which holds whenever the minimum sustainable spread exceeds the per-unit fixed cost threshold. The zero-profit free-entry condition yields equilibrium population:

$$n^* = S\cdot V/F$$

Proposition 3 (Emergence Sufficiency --- Theorem). Given Conditions 1--4, $V_0 > 0$, and $S\cdot V_0 > F$, the intermediary population converges to $n^* > 0$ from any initial state.

Proof. Assumptions: A1 --- $V(0) = V_0 > 0$ (entailed by Condition 2); A2 --- $V(n) = V_0 + \gamma n$, $\gamma \geq 0$; A3 --- $dn/dt = \mu\cdot\pi(n)$, $\mu > 0$, $\pi(n) = S\cdot V(n)/n - F$; A4 --- strategy transmissibility via Lemma 1. Part 1 (Existence): $n^* = S\cdot V_0/(F - S\gamma) > 0$ given $S\cdot V_0 > F$ and $F > S\gamma$; when $F \leq S\gamma$, positive emergence holds a fortiori. Part 2 (Convergence): $W(n) = ½(n - n^*)^2$ satisfies $W(n^*) = 0$ and $W(n) > 0$ for $n \neq n^*$. Define $h(n) = S\cdot V_0/n + S\gamma - F$, strictly decreasing with $h(n^*) = 0$. Then $dW/dt = (n - n^*)\cdot\mu\cdot h(n) < 0$ for all $n \neq n^*$. By Lyapunov's direct method, n^* is asymptotically stable from any $n(0) > 0$. Part 3 (Boundary): At $n \to 0^+$, $\pi(0^+) \to +\infty$ since $V_0 > 0$, making $n = 0$ unstable. Any $\varepsilon > 0$ entry triggers growth toward n^*. Part 4 (Discovery): By Lemma 1, the cost differential is observable before deliberate optimisation, providing the informational precondition for $n(0) > 0$ from the conditions alone. Proposition 3 is proved. ▌

Assumption (Entrepreneurship). Observable profitable opportunities attract entry: any potential entrant who observes $S\cdot V > F$ will enter. This is the standard free-entry assumption in competitive equilibrium models. Lemma 1 proves the spread is observable; this assumption bridges observability to action. It does not require simultaneous observation by all potential entrants --- it requires that the population of potential entrants includes actors who monitor profitable niches and respond to them within finite time.

Lemma 1 (Self-Demonstrating Spread). In any market satisfying Conditions 1--4 with $V_0 > 0$, the cost differential between optimised and unoptimised behaviour is observable before any agent consciously identifies it. Proof. By A1, $V_0 > 0$ implies participants are already

active before any intermediary enters. By Condition 2, they exercise discretion over intensity, timing, and volume, producing nonzero behavioural variation. By Definition 2, $C(x)$ is publicly visible; applied to a population with nonzero variation, it produces cost outcomes with positive variance. Some participants therefore achieve lower expected costs through timing variation alone, before deliberate optimisation. These differentials are observable by construction since $C(x)$ is public. ▌

Counter-cyclical property. Assumption (Inelastic Core): the participant population contains a subset of measure $V_{min} > 0$ whose demand is inelastic with respect to cost --- actors who must transact regardless of the cost rate. This is satisfied whenever the system serves participants with time-critical or obligation-driven needs. Under this assumption: under stress conditions, $V(t)$ rises (inelastic actors must transact regardless of cost) and $E[C(q_{unopt})]$ rises, widening $S(t)$. Therefore $n^*(t) \geq n^*(0)$ during genuine stress --- the equilibrium intermediary population rises under stress rather than falling. Geometric selection follows: when a dominant intermediary collapses, the spread widens and entry rate accelerates. The geometry advertises for replacements at maximum volume precisely when the niche is most open.

3.3 Leaching Impossibility

Proposition 4 (Leaching Impossibility). In any domain satisfying Conditions 1--4 with economic entry, no intermediary can sustain extraction above competitive margins indefinitely.

Proof. Let $\pi(t)$ denote extraction above F. If $\pi(t) > 0$ is sustained, $S \cdot V(t) > F$ holds, triggering entry by Proposition 3. Entry drives $\pi(t) \to 0$. Therefore sustained extraction requires either (a) self-termination by competition, or (b) non-economic barriers --- which is precisely Definition 7. In neither case does indefinite above-competitive extraction persist without non-economic intervention. A stronger characterisation via dynamic limit pricing: let θ^* denote the most efficient latent competitor with sunk entry cost $F(\theta^*)$ and per-unit operational cost $c_v(\theta^*)$. The latent competitor enters if $(V/2r)(S - c_v(\theta^*)) > F(\theta^*)$, where r is the discount rate and V/2 is the post-entry volume share. To prevent entry, the incumbent is constrained to $S \leq 2rF(\theta^*)/V + c_v(\theta^*)$. Total incumbent profit decomposes as $\Pi_i = [c_v(\theta^*) - c_v(\theta_i)]V + [S \cdot V - c_v(\theta^*)V - rF(\theta_i)]$: the first term is Ricardian rent from operational superiority, the second is the strictly bounded equilibrium gap. As $V \to \infty$, the limit pricing bound $S \to c_v(\theta^*)$ and extraction above operational cost collapses to zero.

Leaching is not merely transient under free entry; its maximum magnitude is geometrically capped by the structural parameters of the domain. ▌

The advertising budget formalises this: define $A(t) = S(t) - F/V(t)$ as excess spread above the entry threshold. Every unit of rent extracted above F is a unit of A(t) advertising to competitors. The extraction and the advertisement are the same quantity. There is no position in which rent is extracted and the advertisement is silent. One qualification on observability: A(t) measures extraction above competitive margin as realised in the intermediary's economic position, which may be partially captured internally (bonuses, retained earnings, non-passed-through margin) rather than expressed in observable pricing. Lemma 1 establishes observability of the cost differential between optimised and unoptimised participant behaviour; the advertising budget A(t) aligns with this observable spread when intermediary books are transparent and diverges when internal capture absorbs extraction without price signal. For domains with public pricing (open-order-book markets, statutory deposit returns, regulated fee structures) the divergence is small. For domains with opaque intermediary books the divergence can be substantial, and Proposition 4's entry dynamics are triggered only by the observable component. The full extraction remains geometrically unsustainable under free entry, but the entry signal operates with a lag set by the opacity of internal capture. This lag is finite --- audited financial statements, regulatory disclosure, and competitive intelligence eventually surface internal rent --- but it extends the period during which above-competitive extraction can persist before triggering replacement dynamics. Corollary (Zero-Net-Extraction at Equilibrium). At the steady state of a closed convex loop satisfying Conditions 1--4 under free entry, $A(t) = 0$: no surplus exits the system. Proof: at the zero-profit equilibrium of Proposition 3, each intermediary earns exactly F. By Proposition 4, any surplus above F triggers entry dynamics that drive it to zero. By Condition 4, all non-intermediary revenue returns through Φ to the restoration operator Ψ. Therefore at equilibrium: intermediaries earn F (operating costs, zero economic profit), restoration is fully funded, and no revenue exits to external parties. The mechanism funds itself and produces its output; there is no residual for extraction. This is not a normative result. It is what Propositions 3, 4, and C4 produce when followed to their conclusion. Every system where surplus currently exits --- as dividends, bonuses, or distributions funded by loop revenue --- is in a pre-equilibrium state that the geometry is working to unwind on a timescale determined by α, k, and entry barriers. Equilibrium gap characterisation. The zero-net-extraction result does not claim the mechanism is costless. The intermediary class earns F: this is the mechanism's operating cost, the irreducible residual that sustains the niche. At the zero-profit equilibrium, $S^* = F/V^*$ where V^* is the equilibrium volume. The residual deviation from perfect efficiency is $x_gap = \tau^{-1}(S^*) = \tau^{-1}(F/V^*)$: the deviation level at

which the cost exactly equals the competitive spread. Under the parametric cost function $C(x) = c|x|^\alpha$, this gives $x_{gap} = (F/(cV^*))^{(1/\alpha)}$. The welfare cost of the mechanism is the cost imposed on participants at this residual deviation: $W_{gap} = C(x_{gap})\cdot V^* = F$. The welfare cost of the mechanism equals its operating cost, which equals the intermediary's earnings at equilibrium. Compare this to the open-loop alternative: under open-loop governance without funded restoration, accumulated deviation produces welfare loss $W_{open} = C(\delta)\cdot V$ where δ is the drift under unfunded disturbances. From §3.1b, $\delta \sim (\sigma^2/(2k_F))^{(1/(\alpha+1))}$ under $F(x)$ alone. The mechanism is welfare-improving whenever $W_{gap} < W_{open}$, i.e. whenever $F < C(\delta)\cdot V = c\cdot(\sigma^2/(2k_F))^{(\alpha/(\alpha+1))}\cdot V$. This holds whenever the fixed intermediary cost F is less than the expected open-loop welfare loss from unfunded drift --- a condition that will be satisfied in any domain where disturbances are non-trivial and intermediary infrastructure is not catastrophically expensive. The mechanism is welfare-improving precisely in the domains where it is relevant.

3.4 Structural Alignment vs Incentive Design

The intermediary's alignment is structural, not contingent. Conditional alignment (incentive programmes) weakens when programmes end. Punitive alignment (staking requirements) fails when expected deviation gain exceeds staked value. Structural alignment holds because the geometry makes correct behaviour the only sustainable margin. The attack surface disappears not because enforcement closes it but because the geometry makes it structurally absent.

This distinction --- forced geometry versus incentive design --- changes the competitive landscape. In a standard market, perfection enables capture. In the closed convex loop, perfection is self-defeating: if an intermediary smooths all deviations, the spread disappears and the business model collapses. The geometry enforces a conservative equilibrium without agreement or regulation. The same geometry that produces structural alignment produces structural competition preservation: monopoly is self-dispelling wherever entry barriers are economic. Structural alignment is therefore the evolutionarily stable outcome (ESS, defined formally in §3.5 below): the geometry selects for intermediaries whose survival strategy is identical to the system's health maintenance requirement, and selects against every other strategy. The evolutionary game is formalised in §3.5.

3.5 The Evolutionary Game

The closed convex loop has a natural formulation as an evolutionary game. Each intermediary chooses optimisation intensity $\theta \in [0,1]$. Fitness is $w(\theta) = \pi(\theta)/F$: below 1 the intermediary exits; above 1 it attracts imitators. At $\theta = 0$ (no optimisation): fitness = −1. At $\theta = 1$ (perfect optimisation): the spread disappears and fitness = −1. The fitness landscape has a unique interior peak at θ^* --- the competitive equilibrium where each intermediary earns exactly F. The equilibrium concept is evolutionary stability (Maynard Smith 1982): not which strategy maximises payoff but which survives competition. Intermediaries play to stay in the game, not to win it.

Proposition 7 (ESS = Competitive Equilibrium). The unique evolutionarily stable strategy is θ^*, the competitive equilibrium. No cheating strategy can invade. Proof: an over-extracting mutant generates $A(t) = S(t) - F/V > 0$, the advertising budget that recruits its own replacement via Proposition 3; competition drives its margin to F. An over-optimising mutant drives $S(t)\cdot V < F$ and exits. The unique ESS is θ^*. ▮ The geometry creates a cheating invariant: any above-competitive extraction automatically funds the entry signal that makes it unsustainable. Cheating does not get caught. It recruits its own replacement. The ESS framing connects the earlier results: Proposition 3 establishes convergence to n^* (the ESS population); Proposition 4 is the impossibility of sustained cheating under economic entry; the captured loop is the ESS of a mechanism aimed at S_c rather than S^*. Designer selection operates at second order by the same logic (§7.3).

Proposition 5 (Geometric Dominance). Consider a non-atomic continuum of agents where agent i has private benefit $b_i \sim F$, chooses $a_i \in [0,1]$, and receives payoff $\pi_i(a_i, x) = b_i\cdot a_i - \tau(x)\cdot a_i$, where $V_i(a_i) = b_i\cdot a_i$ depends only on a_i and $C_i(a_i, x) = c\cdot|x|^\alpha\cdot a_i$ is convex in a_i for all $x \geq 0$. Then for every agent i and every aggregate state x, the dominant strategy is $a_i^* = 1\{b_i > c\cdot|x|^\alpha\}$. Strategic interaction does not alter this strategy. The aggregate dynamics that follow from individual dominant strategy play coincide with $dx/dt = -k\cdot c\cdot|x|^\alpha\cdot\mathrm{sign}(x)$, established as asymptotically stable by §3.1a. The game collapses to the geometry. Proof: see Appendix B.

Proposition 6 (Necessity of the Four Conditions). A mechanism M produces structural alignment (Definition 5) only if Conditions 1--4 are all satisfied.

Proof. (C1 violation) If no natural equilibrium S^* exists, "correct behaviour" has no geometric anchor. No cost function C(x) can produce convergence to a target that does not exist. (C2 violation) If agents have no discretion, $S := E[C(q_{unopt})] - C(q_{opt}) = 0$. By Proposition 3, $n^* = S\cdot V/F = 0$. No intermediary class emerges. Definition 5 cannot be satisfied. (C3 violation) If deviation imposes no genuine systemic cost, C(x) cannot be calibrated to system state. Any

cost imposed is arbitrary extraction; the restoration operator Ψ has no meaningful target. The structural connection of Definition 5 is severed. (C4 violation) The permanence requirement of Definition 5 is not a design choice but a consequence of the paper's central question: does the mechanism still produce alignment if everyone administering it disappears? A mechanism whose alignment holds only while external conditions cooperate is a managed system. Structural alignment requires permanence by definition --- independently of C4 --- because permanence is what separates mechanisms from policies. C4's necessity then follows without circularity: it is what makes the independently-motivated permanence property hold. Intermediaries may emerge without C4 (Proposition 3 requires only Conditions 1--3 plus S·V \> F). C4 is not necessary for emergence. C4 is necessary for alignment to be structural rather than contingent. Without C4, the autonomous dynamics F(x) provide some restoring force, but the practical stability bound from §3.1b gives E\[\|x\|\] \~ (σ^2/(2k_F))\^(1/(α+1)). For σ above the domain threshold, long-run deviation collapses the niche: S(t)·V(t) eventually falls below F and n*(t) = 0. Alignment was real but contingent on σ remaining below that threshold --- factors outside the mechanism's design. That is managed dependency, not structural alignment. In each case violation is sufficient to prevent structural alignment. Conditions 1--4 are therefore necessary. ▌

3.7 Timescale Commensurability and Effective Restoration Rate

The deterministic Lyapunov proof of §3.1a and the stochastic confinement bound of §3.1b assume continuous-time dynamics in which the mechanism's measurement of R(t) and its restoration response through Φ and Ψ operate at timescales commensurable with the fastest extraction timescale in the domain. Real mechanisms measure and respond in discrete cycles. When the extraction timescale is substantially faster than the response cycle, the effective restoration rate at the extraction timescale is not k but a reduced quantity k_effective.

Definition 10 (Effective Restoration Rate). Let τ_extract denote the characteristic timescale of the fastest extraction behaviour in the domain and τ_response the characteristic timescale of the measurement-plus-restoration cycle. The effective restoration rate at the extraction timescale is k_effective ≈ (τ_extract / τ_response) · k for τ_extract ≤ τ_response, with k_effective approaching zero as τ_extract / τ_response approaches zero.

Proposition 10 (Timescale Commensurability). Structural alignment requires k_effective ≥ k_min at the fastest extraction timescale in the domain, where k_min is the minimum viable restoration rate from §7.2. Proof. The stochastic confinement bound |x_∞| ~ (σ^2/(2k_effective·c))^(1/(α+1)) governs long-run deviation at the extraction timescale. For

k_effective < k_min the bound exceeds $\tau^{-1}(F/V)$, the niche-collapse threshold, and the intermediary class does not sustain entry. The geometry's structural alignment result collapses to contingent alignment dependent on external enforcement at the response timescale. Since k_effective = (τ_extract / τ_response) · k, a mechanism with τ_extract << τ_response can satisfy k > k_min nominally while violating k_effective > k_min operationally. ▮

This formalises a distinct failure mode from capture as defined in §4.2. Capture is stable misalignment toward a surrogate equilibrium under a correctly-aimed geometry. Timescale-mismatch failure is correct aim under geometry that cannot execute at the extraction timescale. The identification criterion differs: capture is diagnosed by intermediary opposition to recalibration; timescale-mismatch failure is diagnosed by persistent extraction at sub-cycle timescales that the measurement apparatus does not resolve. The two failure modes can compound: an intermediary class that extracts at timescales below the response cycle generates A(t) that the mechanism cannot observe, which makes the extraction invisible to the identification criterion that would otherwise detect capture.

Most financial capture operates through this mode. High-frequency trading, flash loans, and intraday leverage cycles have τ_extract in milliseconds to hours; statutory capital, stress testing, and regulatory reporting have τ_response in quarters to years. The ratio τ_extract / τ_response in such cases is of order 10^{-6} to 10^{-8}, making k_effective operationally zero regardless of k_nominal. The paper's interbank application (§5.2) is substantially affected by this failure mode, as noted in that subsection. Domains with cleanly commensurate timescales --- grid frequency measured continuously with sub-second response, deposit-refund with return-rate measurement at the same cadence as container circulation --- do not exhibit this failure mode and are correspondingly cleaner demonstrations of the framework. The prerequisite P1 should be read as the qualitative version of Proposition 10: timescale commensurability is not a binary check but a constraint on k_effective relative to the fastest extraction channel the domain admits.

4. Failure Modes

4.1 The Open Loop

The four prior mechanisms most similar to the closed convex loop each satisfy three of the four conditions and fail at exactly one: Condition 4. The Tobin tax routes revenue to

international development programmes. Pigouvian taxes route revenue to general government funds. AMM fee structures route revenue to liquidity providers rather than the protocol. EIP-1559 burns fees rather than recycling them. In each case the designers understood the loop structure. Closing the loop meant giving up the revenue. They chose extraction. The failure was not incompetence. It was the rational choice of designers operating within a system that rewards extraction.

The complement definition enumerates the full failure set. A domain fails the mechanism if any of the following hold: (a) no aggregate natural equilibrium; (b) no participant discretion; (c) no structural coupling; (d) timescale mismatch between cost signal and equilibrium; (e) unobservable or manipulable state variable; (f) open revenue loop; (g) $S \cdot V < F$ for all feasible intermediary configurations. Every individual-condition failure maps onto exactly one of these seven properties.

4.2 The Captured Loop

The captured loop is not merely a failure mode of the mechanism. It is the strongest available evidence of the mechanism's power. The same geometry that produces structural alignment produces stable misalignment when aimed at the wrong target --- with identical mathematical reliability. This is not a weakness. An amplifier that stabilises whatever target it is given demonstrates precisely the amplification property the paper claims. The captured loop proves the geometry works. The question of whether it is aimed correctly is a separate design problem --- one the geometry itself partially resolves in Tier 1 systems, where entry dynamics select against captured intermediaries. In a correctly designed Tier 1 mechanism, an entrant offering services calibrated to S^* earns higher margins than one calibrated to S_c --- because S^* is the true attractor and deviations from S^* are larger, generating more spread to exploit. Entry therefore selects for correctly aimed intermediaries, making correct aim an emergent competitive property rather than a design requirement. Capture is not instability. It is stability with the wrong aim --- and in Tier 1 systems, wrong aim is competitively disadvantaged.

The rational capturing actor chooses deviation $d^* \in (0, D)$ rather than full capture, because full capture destroys the extraction base.

Formal result (interior optimum). Let $E(d) = r \cdot d \cdot A(d)$ where $r > 0$ and $A(d)$ is system activity with $A(0) > 0$, $A'(d) < 0$, $A(D) = 0$. By IVT, a unique $d^* \in (0, D)$ satisfies the first-order condition $d = -A(d)/A'(d)$. Under log-convexity of $A(d)$ --- satisfied when system activity declines with diminishing marginal sensitivity to deviation pressure, as occurs when

participants near behavioural limits absorb additional deviation with decreasing marginal response --- d* is the unique global maximum. The optimal deviation is where marginal extraction gain and marginal system cost exactly balance.

The practical consequence: the captured loop self-limits at d* rather than tending toward full capture. A self-limiting captured loop is more durable, not less dangerous, than one that destroys its host. This is precisely the condition that makes the identification criterion --- intermediary class opposing recalibration --- the correct diagnostic. This is the structural interpretation the framework offers for a class of institutional failure. SVB, Signature Bank, LTCM, and Terra/Luna are candidate cases where the framework's predicted pattern --- intermediary margins sustained by routing surplus away from the substrate that makes those margins possible, with governance extending T_collapse rather than closing the loop --- is consistent with the documented failure sequence. The framework does not in this paper trace the specific mechanisms (interest rate risk, HTM accounting, collateral cascades) through which each collapse occurred. The claim here is the structural one: mechanisms that violate Condition 4 have finite T_collapse bounded by their extraction rate relative to substrate regeneration, and governance applied to such mechanisms extends rather than eliminates that countdown. Whether any specific institutional failure fits this pattern is a case-by-case empirical question the structural audit (§7.3a, under the qualifications noted there) takes up but does not resolve. The theoretical claim stands independent of the specific illustrations: where the four conditions are not satisfied, the stability properties of Propositions 1 and 10 do not hold, and the institutional apparatus that maintains the mechanism is evidence of what the geometry fails to accomplish on its own. A clarification on scope: while the d† threshold characterises when a captured loop is preferable to open-loop governance in Tier 2 systems, any non-zero capture in a Tier 1 candidate is an unstable transition state rather than an acceptable equilibrium. The mechanism's own geometry works to dismantle it: the moment a surrogate equilibrium begins to form, the spread widens, advertising for competitors who will eat the capture margin via the zero-profit free-entry condition of Proposition 3. Capture is structurally self-defeating in a correctly designed Tier 1 mechanism. The d† analysis applies to Tier 2 systems deciding whether to recalibrate --- not to Tier 1 candidates where capture represents a design failure.

4.2a The Capture Threshold d†

The captured loop is not purely harmful relative to the open loop. A lightly captured system (S_c close to S*) is better than no mechanism at all. Define:

C_cap(d) ≈ k·d\^α (social cost of captured loop at deviation d)

C_open (social cost of open-loop governance, approximately constant)

R_corr(d) (one-time correction cost, increasing in d)

The crossover threshold satisfies:

d† = (C_open / k)\^(1/α)

For d \< d†, the captured loop is preferable to open-loop governance in steady state. For d \> d†, open-loop governance is preferable. Three implications: (1) higher α reduces d† --- high-convexity domains tolerate less capture; (2) higher C_open increases d† --- expensive governance tolerates more capture; (3) time-to-switch after d exceeds d† is T_switch = R_corr(d) / \[C_cap(d) − C_open\]. Strong opposition to recalibration is evidence that d is large --- precisely the conditions under which recalibration authority must be structurally protected.

5. Domain Applications

The formal results of §3 establish the mechanism's stability properties, emergence conditions, and failure modes in general terms. The following applications translate those results into structural arguments for two principal domains, preceded by the motivating observation that produced the framework. The applications demonstrate that the conditions are plausibly satisfied and identify the intermediary class the mechanism predicts. The domains were chosen for architectural and institutional independence from each other and from the protocol in which the principle was first observed. The analyses are structural rather than empirical: we argue the conditions are plausibly satisfied, not that they have been quantitatively confirmed. Each domain analysis is a Tier 3 → Tier 2 transition proposal: the current system is Tier 3 (continuous human correction required for rate-setting and threshold management); the closed convex loop installs a self-executing inner loop inside the existing managed shell.

5.1 DeFi Settlement Architecture (Motivating Observation)

The principle was first observed in a DeFi settlement protocol where a convex tax on short-term trading volume, with revenue routed to the reserve, produced a measurable behavioural shift toward smoothed transaction timing before any external communication about the design. No intermediary was recruited. This observation motivated the framework. The DeFi domain satisfies all four conditions in principle: reserve health ratio as natural

equilibrium (Condition 1), participant discretion over transaction timing (Condition 2), reserve destabilisation as genuine systemic cost (Condition 3), and convex tax revenue routing to the reserve (Condition 4). The two principal domain demonstrations follow.

5.2 Interbank Lending Markets (Candidate Domain with R* Contamination)

Interbank lending is presented here as a candidate domain rather than a cleanly demonstrated application. Condition 1: systemic leverage level L_0 as natural equilibrium --- revealed inductively from historical data. Condition 2: institutions exercise discretion over leverage timing and counterparty selection. Condition 3: overleveraging imposes genuine systemic cost via contagion. Condition 4: convex systemic risk charge with revenue routing to a Systemic Liquidity Reserve. Three limitations reduce the strength of the application. First, R* sits in Regime B: historically calibrated rather than physics-defined, introducing contamination risk. The F(x) restoring dynamics required by Definition 1 are read from data generated under the prior governance regime; removing that regime may alter F(x) itself, and the framework does not currently resolve this identification problem. Second, with O(10--50) major participants, strategic interaction is first-order and the continuum approximation of Appendix B breaks down; the finite-N extension identified as future work (§7.4) is needed before the formal machinery applies cleanly. Third, timescale commensurability (see §3.7 and Proposition 10) is contested: extraction can occur at millisecond timescales while regulatory measurement and response operate at quarterly timescales, producing a k_effective substantially below k_nominal. The interbank case is therefore illustrative of how the framework would apply if the identification, finite-N, and timescale issues were resolved. It is not a headline demonstration. The structural audit (§7.3a) documents a pattern of Condition 4 violation in financial institution failures that is consistent with the framework's predictions under the qualifications above, but this pattern is vocabulary validation rather than structural identification.

5.3 Electrical Grid Load-Balancing

Condition 1: grid frequency at 60 Hz (physically defined). Condition 2: industrial consumers exercise genuine discretion over consumption timing. Condition 3: frequency deviation imposes genuine systemic cost. Condition 4: a convex deviation charge during periods of frequency imbalance, with revenue routing to grid stabilisation infrastructure.

This is the cleanest Tier 1 candidate among the three: R* is physically defined with no contamination risk, state variable is continuously measurable, and the intermediary class

(demand-response aggregators) already exists in nascent form. Demand-response aggregators --- firms that earn by optimising industrial consumption timing in response to grid frequency signals --- have emerged in deregulated electricity markets across the US, UK, and Australia. Their emergence is consistent with Proposition 3's prediction that where the four conditions are satisfied an intermediary class emerges without design, but this is observational consistency rather than causal identification: these aggregators arose from deregulated wholesale markets and capacity payment schemes, and the four-condition mechanism may be one of several sufficient causes. Their business model is structurally dependent on frequency deviation persisting --- a perfectly balanced grid eliminates their margin --- which is consistent with the self-dispelling monopoly property of the mechanism. The governance corollary holds at full strength.

5.4 The Linux Kernel: Defection-Dominance and the Unkillable Equilibrium

The preceding domain applications treat the closed convex loop as a proposed design. This subsection treats a different class of evidence --- a case where the mechanism is already operating in a domain nobody designed it into, where the intermediary class emerged without recruitment over three decades, and where the geometry has produced a property the formal framework predicts but has not previously named: defection-dominance under adversarial entry. The Linux kernel and its surrounding ecosystem provide the cleanest extant instantiation of a Tier 1 mechanism. It was not designed as one. It became one through the interaction of structural properties whose conjunction satisfies the four conditions.

5.4.1 Condition Mapping

Condition 1 (natural equilibrium): the set of operating-system configurations that run the world's production software infrastructure. This attractor is revealed by the aggregate behaviour of every workload running on general-purpose computing, and converges on the kernel interfaces, subsystem behaviours, and performance characteristics that production workloads require. S* is the kernel that supports the workloads. Condition 2 (participant discretion): contributing organisations exercise full discretion over what they contribute, when, and at what intensity. Condition 3 (genuine systemic cost): a regression in a kernel subsystem imposes direct cost on every contributor whose product depends on it, measurable in engineering time, customer incidents, and competitive position. Condition 4 (loop closure): the GPLv2 license is loop closure written into statute. Revenue --- in the form of engineering output --- cannot permanently exit the domain while the kernel is being used as the kernel. The licensing structure forces derivative work back into the commons as the

cost of distribution. The restoration operator is the kernel development process; the revenue-routing map is the license.

5.4.2 The Intermediary Class

The dominant contributors by volume and subsystem ownership are Intel, Red Hat (IBM), Google, Meta, AMD, Huawei, Samsung, SUSE, and Oracle --- corporations whose products require the kernel to function correctly, whose competitive position depends on specific subsystems working for their hardware or their workloads, and for whom the marginal cost of maintaining the subsystems they depend on is strictly lower than the marginal cost of not maintaining them. This is Definition 5 satisfied directly. Each contributor's profit motive is structurally identical to the kernel's health --- not because the contributor values the kernel but because the geometry makes correct behaviour the only sustainable margin. Proposition 3 predicts this emergence without design. The emergence occurred without design. The composition of the class (hardware vendors contributing to hardware subsystems, hyperscalers contributing to scalability, distributors contributing to stability) corresponds precisely to the cost-differential exploitation predicted by Proposition 3.

5.4.3 Defection-Dominance

The Linux case exhibits a property the framework has thus far stated only implicitly: the equilibrium is not merely stable against perturbation --- it is stable against deliberate attack by actors who would prefer it did not exist.

Definition 8 (Defection-Dominance). A closed convex loop satisfying Conditions 1--4 under free entry exhibits defection-dominance if, for any actor A with the capability to produce an equivalent instantiation, the strategy of abandoning or suppressing the mechanism is strictly dominated by the strategy of participating in it, regardless of the strategies of other actors.

Proposition 8 (Adversarial Entry Produces Replacement). In any domain satisfying Conditions 1--4 where the set of potential providers contains more than one actor with the capability to produce an equivalent instantiation, no single actor can durably suppress the mechanism. Suppression by actor A opens a spread $S(A)$ equal to the cost differential between actors using the mechanism and actor A's substitute; by Proposition 3, any potential entrant observing $S(A) > F$ will enter and capture the displaced volume. Suppression is therefore a unilateral handicap transferring market share to competitors. Collective suppression by all capable actors is not a Nash equilibrium: the first defector captures the entire displaced volume, which dominates honouring the suppression for every actor in isolation. The only Nash equilibrium is universal participation.

Proof sketch. Let N actors have the capability to produce instantiations, with current participation producing per-actor margin m. Suppose actor A suppresses its participation. The volume V(A) previously served by A becomes available to entrants. The spread exceeds F (since A's prior participation was sustainable). By Proposition 3, entry occurs and convergence to the zero-profit equilibrium resumes. A has shed volume to competitors and gained nothing. For collective suppression: the first defector observes spread times total volume exceeding F by a factor of N and captures the entire displaced market. Defection from the suppression coalition is strictly dominant for every member. The coalition is not a Nash equilibrium under any N >= 2.

5.4.4 Rent-Extraction Symmetry

Defection-dominance has an interpretation as rent-extraction symmetry. In a standard extractive system, rent is captured by the actor controlling access to a scarce resource. Under the closed convex loop with free entry, any actor suppressing access to the mechanism creates a scarce resource --- participation through an alternative provider --- whose rent accrues to the first actor to provide it. The suppression converts the suppressor's volume into a capturable opportunity for competitors. The rent that the suppression was intended to transfer to a proprietary substitute is captured instead by competitors the suppressor cannot reach. This is not hypothetical: it is the direct consequence of the dominant strategy result.

5.4.5 Scope and Capture

Defection-dominance requires that the set of potential providers contain more than one actor with equivalent capability. The property holds at civilisational scale and fails at scales where the capability gap between actors is large enough that no credible substitute provider exists. The claim is: no single actor or coordinated coalition in a domain containing multiple capable providers can durably suppress the mechanism. It is not: the mechanism cannot be suppressed under any conditions.

Linux has not been suppressed. It has also not, at the time of writing, been captured in the sense of Definition 7 --- but the framework's own predictions indicate that capture is the more likely failure mode than suppression. The concentration of kernel maintainer authority among employees of a small number of hyperscalers, the shaping of subsystem decisions around workloads specific to those hyperscalers, and the marginalisation of use cases that do not serve the dominant contributors are all consistent with the early stages of capture toward a surrogate equilibrium S_c. The geometry that makes Linux unkillable from outside does not make it incorruptible from within.

5.4.6 Implications for the Framework

The Linux case provides three contributions beyond demonstrating that the four conditions are satisfied in a non-designed system. First, it establishes that loop closure can be achieved through licensing rather than revenue routing --- the GPL is Condition 4 operating on engineering output rather than monetary revenue. This suggests that Condition 4 is better understood as a constraint on any form of value that exits the domain, not merely money. Second, it demonstrates that defection-dominance is a derivable property of the mechanism under free entry with multiple capable providers (Proposition 8), strengthening the stability result of Proposition 1. Third, it clarifies what Tier 1 means in practice. A Tier 1 mechanism is not one whose managed shell has been eliminated --- Linux operates within international copyright enforcement, export controls, and commercial infrastructure. A Tier 1 mechanism is one whose inner loop is self-enforcing and whose boundary conditions would, if attacked, produce dominant-strategy defection from the attack rather than compliance with it.

6. The Governance Corollary

Governance in an economic system performs three functions: (1) rate-setting --- pricing a key variable in real time; (2) threshold management --- preventing the system from crossing structural limits; (3) emergency response --- handling events outside the normal operating envelope. A correctly designed closed convex loop eliminates Functions 1 and 2 within the operating domain D. Function 3 remains necessary because no finite mathematical system anticipates every event. The scope of governance is not abolished; it is precisely narrowed to emergency response and periodic recalibration of the cost schedule parameters (α, τ_{max}, R^*).

The timing inequality formalises the rate-setting advantage. Let τ_{event} denote stress onset, τ_{detect} detection latency, $\tau_{response}$ time to corrective effect. Committee governance is bounded below by reporting cycles, deliberation, and implementation. An automated mechanism keyed to live state variables has latency approaching the state update time --- orders of magnitude faster. A committee setting rates 48 hours after stress onset is calibrating to conditions that no longer exist. The convex cost function calibrates continuously to the condition that exists now. A publicly known cost schedule additionally constitutes endogenous forward guidance: participants forecast their own costs from $C(x)$ without committee announcements. The mechanism absorbs the contemporaneous price-signal channel of rate-setting and the expectations channel wherever the cost schedule provides

sufficient information about future costs. What it cannot replace is novel-risk coordination --- regime changes, tail events, structural shifts outside the cost schedule's parameterisation.

The corollary holds at full strength in Regime A domains (grid frequency: physically defined S*, continuous state measurement). It holds in direction but with attenuated strength in Regime B domains (interbank: historically calibrated R*, off-chain measurement). Authority over target recalibration must be structurally separated from those whose margin depends on the current deviation metric, because captured equilibria actively resist correction. The identification criterion of Definition 7 provides the diagnostic.

7. Discussion and Scope

7.1 The Four Conditions and Mechanism Design

The four conditions generalise the mechanism design tradition\'s static constraints to the dynamic setting. Incentive compatibility, individual rationality, and budget balance describe equilibrium properties at a point in time. The present conditions describe what is required for rational behaviour to continuously reproduce the conditions for good outcomes. Condition 4 is the critical addition: it is what mechanism design requires when you ask it to work indefinitely rather than once.

The prior literature was not wrong. It was building machines that needed operators. The four conditions specify the first class of machines that operate themselves --- and identifies the extraction of loop revenue as the precise and consistent reason no one built this before. The consistently violated condition --- Condition 4 --- is precisely the condition whose closure eliminates the revenue stream that existing open-loop intermediaries depend on maintaining. The design principle this implies can be stated simply: build the system so that the only way to profit is by making the system healthier. Not as a normative aspiration. As a structural consequence of the cost geometry. A mechanism with this property does not require governance to survive. It requires governance only when it needs to be introduced --- to set the initial parameters, calibrate R*, and establish the cost schedule. After that, the geometry governs. This is not idealism. It is what the four conditions produce when followed to their conclusion. The formal object corresponding to a closed-loop banking mechanism is specifiable: a system that stores and transfers value, whose intermediary class earns by maintaining the reserve health that makes storage safe, whose surplus returns to the reserve, and whose extraction is geometrically self-defeating under the timescale commensurability

condition of Proposition 10. The claim is an existence result about the formal specification, not a counterfactual about existing institutions. Whether such a mechanism would be adopted, how it would be calibrated, and whether the parameter-setter problem (§7.2) would be solved in its favour are separate questions the framework does not resolve. What the framework does establish is that the specification exists and that any banking mechanism whose cost geometry satisfies the four conditions with adequate k_effective operates under the stability and alignment properties of Propositions 1--10 regardless of the regulatory overlay.

7.2 Scope and Limitations

The proofs establish local asymptotic stability and practical stability under continuous disturbance. Global stability of a single instantiation remains open. A further scope boundary: the stability results hold for k \> 0 constant. In practice k is itself a governance parameter --- whoever controls the restoration operator controls its efficiency --- and k-degradation under adversarial conditions is the mechanism analog of regulatory capture applied to the restoration operator itself. Formally characterising k-degradation dynamics is identified as future work. The existing stochastic confinement bound in §3.1b implies a minimum viable k: setting the confinement bound \|x_inf\| \~ (sigma\^2/(2kc))\^(1/(alpha+1)) equal to the niche-collapse threshold tau_inv(F/V) and solving gives k_min = sigma\^2 / (2c · \[tau_inv(F/V)\]\^(alpha+1)). For k \> k_min the mechanism maintains alignment; for k \< k_min the system's long-run deviation is sufficient to collapse the intermediary niche, transitioning the mechanism toward Tier 2 behaviour. This bound characterises how much restoration efficiency degradation the mechanism can absorb before structural alignment fails. A related corollary concerns extraction timescales. Define T_collapse as the first time E\[\|x(t)\|\] under extraction dynamics exceeds tau_inv(F/V) --- the threshold at which the intermediary niche collapses. From the stochastic framework of §3.1b, T_collapse is finite and decreasing in extraction rate: the faster revenue exits the loop, the sooner the system crosses the niche-collapse threshold. For any extraction rate below the critical level, T_collapse may exceed human lifespan --- the system appears stable from inside any individual observation window. From the geometry it is a countdown. This formalises the observation that extractive strategies can appear successful indefinitely until they do not: the aggressive strategy is not inside the confinement bound longer, it is eating the bound itself, shrinking the basin of stability until a normal-sized perturbation becomes fatal. The conservative strategy --- extraction at or below the level that keeps k above k_min --- has T_collapse approaching infinity. This formalises a design target, not a design guarantee. The mechanism does not require decision-makers to be good. It requires

T_collapse to fall inside the decision-maker's career horizon T_career --- the window during which the consequences of extraction land on the actor who made the extraction decision. Setting T_collapse = T_career and solving for the minimum convexity threshold yields α_min: the value of α above which extraction is geometrically self-punishing within the decision-maker's own tenure. Two qualifications bound this result. First, the cost function $C(x) = c|x|^\alpha$ has derivative $c\alpha|x|^{(\alpha-1)}$ that approaches zero as x approaches zero for $\alpha > 1$. Higher α makes the cost function steeper at large $|x|$ but flatter near equilibrium, which without a positive floor τ_base permits a sophisticated extractor to operate just below the threshold where the cliff begins and extract freely until that threshold is crossed. The α_min result therefore requires τ_base > 0 doing active work: the floor prevents near-equilibrium free extraction, and α controls how rapidly cost escalates with deviation. Both parameters bear on whether extraction is self-punishing within a career horizon; the design answer is (α, τ_base), not α alone. Second, and more consequentially, α_min characterises the design target but does not specify how the target is hit. The decision-maker whose T_career bounds the relevant feedback is not the intermediary --- whose behaviour the geometry governs in real time --- but the parameter-setter who calibrates α, τ_max, τ_base, k, and R*. These parameters are set outside the closed loop, by actors whose career incentives this framework does not model. The parameter-setter faces their own T_career versus T_collapse calculation about the mechanism they are calibrating, and nothing in the current framework ensures that calculation produces $\alpha \geq \alpha$_min. Designer selection as second-order evolutionary stability (identified in §7.4 as future work) is the missing result. α_min is therefore the correct design target, the condition for geometric self-punishment within a given horizon, and the characterisation of what convexity must achieve --- but it is not a claim that the target is met. It is a claim about what meeting the target would require. The design question \"how convex does the cost function need to be?\" now has a concrete answer in terms of α, k, and the career horizon of the actors whose behaviour the mechanism is meant to govern. T_collapse is not a scope limitation of the theory. It is the central result stated in time. Every extractive mechanism has one. The governance overhead, the regulatory capital requirements, the central bank backstops --- these extend T_collapse. They do not eliminate it. A mechanism bonded to its equilibrium, satisfying all four conditions, has T_collapse approaching infinity. One that is not has a countdown running from the day it was built. The theory does not say extraction is wrong. It says extraction is mortal. And the mortality is not optional, not a matter of governance quality, not fixable by better management. It is a consequence of the cost geometry. The globally persistent property --- that no finite disruption permanently prevents the mechanism from being reinstantiated --- is established by Proposition 3 and the counter-cyclical spread property and does not require global stability of a single instance.

The paper proves convergence to S* and the properties of that convergence. It does not establish that S* is welfare-optimal relative to alternative targets. In Regime A domains, S* is physically defined and welfare assessment is largely tautological --- grid frequency at 60 Hz is not a welfare choice. In Regime B domains, S* is historically calibrated and there is no guarantee the natural equilibrium is welfare-optimal: credit markets may have natural equilibria that include periodic crises, and stabilising around such an equilibrium may not be desirable. The mechanism is alignment machinery conditional on target quality. Target quality is a prior, domain-specific determination that this paper treats as given. Addressing whether S* is the right target, and not merely where the geometry points, requires welfare analysis that is outside this paper's scope and would constitute a separate contribution.

R* contamination is a genuine limitation in Tier 2 domains: the historical data from which R* is read has been shaped by prior managed systems. A designer reading R* from that data may calibrate toward a historically produced S_c rather than the structural equilibrium. The evolutionary deployment model --- finding S* across multiple generations of deployment --- is therefore the more accurate description of how the mechanism operates in practice. A further scope boundary: in domains with multiple legitimate equilibria, the mechanism is a stabiliser rather than an optimiser. The geometry stabilises whichever basin of attraction the system starts in with high reliability; the choice of which equilibrium to aim for is a governance problem the geometry cannot resolve. This is distinct from the captured loop (which is a wrong target) --- it is the case where several targets are each structurally defensible and the designer must choose among them. In such domains the paper's contribution is the alignment machinery conditional on a target, not the identification of the target itself.

Simulation cannot verify the theory\'s infinite-time claims and simulation by the author cannot avoid contamination from design choices. The appropriate verification is independent observation of whether the predicted intermediary class emerges in domains satisfying the four conditions, conducted by researchers without access to the theoretical framework. The theory can be falsified at every failure point: every time a system violates Condition 4 and does not collapse on a predictable timeline, every time a captured loop is identified and the intermediary class does not oppose recalibration, every time Conditions 1--4 are satisfied and intermediary emergence fails to occur. A companion paper (Buday, 2026) provides exhaustive classification of sixty systems across three institutionally independent domain categories --- DeFi protocol collapses (2020--2025), systemically important financial institution failures (1980--2025), and historical commons and infrastructure failures --- under the four conditions, finding zero false negatives and failure mode correspondence in all sixty

cases. The theory paper stands on its formal contributions alone; the classification paper provides the diagnostic falsification exercise. What cannot currently be tested is the large positive confirmation. The distinction: the theory can be falsified at every failure point but can only be confirmed through deployment that the current institutional landscape structurally prevents. The theory's epistemological standard is engineering, not prophecy: it characterises failure modes, establishes margins of safety through the stochastic confinement bound, and provides a diagnostic for detecting capture --- not a guarantee of permanent stability, which is unavailable for any finite engineered system. A bridge's structural analysis tells you why it fails and what margins prevent failure; it cannot tell you the bridge will stand forever. The four conditions, the captured loop diagnostic, and the T_collapse formalism provide the same service for mechanism design: here are the failure modes, here is the margin of safety, here is what to monitor and when to worry. On the game-theoretic concern: Proposition 5 (Geometric Dominance, §3.4) formally establishes that the aggregate dynamics are derived from individual dominant strategy behaviour in a non-atomic continuum game with heterogeneous private types --- the game collapses to the geometry, and Appendix B provides the proof.

7.2a The Observer Problem at Planetary Scale

The framework predicts T_collapse as a finite countdown for every extractive mechanism. At small scales T_collapse is observable: banks fail, protocols collapse, fisheries crash. At large scales T_collapse becomes commensurate with or longer than the observer's own existence, and the framework's predictions become structurally unfalsifiable from inside the system they describe. A civilisation running a mechanism with T_collapse of 100,000 years cannot, at any point during its operation, distinguish that mechanism from a closed-loop one. This is different from the acknowledgement above that the theory cannot currently be positively confirmed --- that is about the current institutional environment. The observer problem is about the framework predicting a regime in which its predictions are in principle unobservable by the system the predictions concern.

Stating this makes the paper more honest rather than less. It demonstrates that the framework knows where its own predictions cease to be evaluable and why. A theory that predicts unfalsifiability at certain scales is stronger when it names that limit than when the limit is identified externally as a weakness. The framework's epistemic reach is bounded by the timescale of its predictions relative to the timescale of its observers, and this bound is itself a derivable consequence of the framework rather than an external limitation imposed upon it.

7.2b The Bootstrapping Problem

The framework specifies what mechanisms look like once they exist, how they converge, how they persist, and how they fail. It does not specify how Tier 1 mechanisms come into existence in a world dominated by Tier 3 mechanisms whose actors control the regulatory environment. Proposition 3 describes emergence within a domain that already satisfies Conditions 1--4. It does not describe how a domain comes to satisfy those conditions in the first place. The DeFi motivating observation works because the blockchain substrate existed; Linux works because Unix and the GPL existed; deposit-refund works because national regulatory authority existed. None of these founding conditions is produced by the mechanism itself.

Without this acknowledgement, the paper reads as if building closed-loop mechanisms is primarily a design problem. The harder problem is political: the designers do not currently have the authority, and the actors who do have authority are the ones whose margins depend on the current extractive geometry persisting. The framework is a specification of what to build; it is not a theory of how to acquire the authority to build it. This clarifies the paper's theory-of-change and pairs with the Tier 2 transitional framing: Tier 2 is the correct first move partly because it does not require bootstrapping a Tier 1 mechanism from scratch within a hostile institutional environment. The inner loop demonstrates its properties while the managed shell provides political cover.

7.3 Empirical Evidence

The propositions proved in Sections 3--6 are formal results about asymptotic behaviour. Asymptotic properties cannot be directly observed in finite data. This section therefore does not claim to prove the theory. It presents three independent bodies of evidence --- a structural audit, a cross-country test in electricity markets, and a mechanism case study in deposit-refund systems --- that are individually insufficient but collectively establish plausibility by the standard appropriate to formal economic theory with asymptotic claims. General equilibrium theory predicts convergence to Pareto-efficient allocations but no dataset has confirmed this to completion; it is accepted because its predictions hold where testable and fail where conditions are violated. The evidence below follows the same structure.

7.3a Structural Audit: Failure-Mode Correspondence

The companion paper (Buday, 2026b) examines 60 system failures across three domains: decentralised finance (20 cases), financial institutions (20 cases), and commons or

infrastructure systems (20 cases). Each system is assessed against the four conditions and the failure mode is classified by the condition(s) violated. The audit, as author-classified, finds no system satisfying all four conditions that experienced the type of structural failure predicted by the framework, and 49 of 60 failing cases (81.7%) map to Condition 4 violation as the primary failure mode. Failure-mode correspondence holds across categories: C1 violations produce equilibrium misidentification; C2 violations produce coerced-participation failures; C3 violations produce cost-externalisation failures; C4 violations produce intermediary capture or extraction. Two qualifications are essential. First, the classification was performed by the author without pre-registered protocol or independent coder verification. Until independent coding is conducted, the audit establishes diagnostic vocabulary rather than empirical test: it shows that the four-condition framework provides a consistent post-hoc classification of why these systems failed, not that the framework uniquely predicts their failure against alternative classifiers. Second, the framework's interpretive flexibility --- Remark 1 (passive closure via biological regeneration), the ability to select which loop is examined in multi-loop systems --- is structural rather than ad hoc, but it is flexibility nonetheless, and its effect on classification uniqueness has not been independently tested. Independent coding under a pre-registered protocol is registered as a priority empirical project. The audit's current contribution is vocabulary validation: the framework correctly identifies candidate failure modes and their correspondence to the conditions. It does not establish that satisfying all conditions produces permanent alignment --- that claim remains a formal result whose empirical status is inherently provisional --- and it does not establish that the framework's classification outperforms alternative classifiers on the same cases, which is the test independent coding would provide.

7.3b Cross-Country Test: Electricity Balancing Markets

The companion empirical paper (Buday, 2026c) tests whether Condition 4 --- operationalised as regulatory market openness to demand-side flexibility --- predicts demand-response aggregator emergence across 30 European electricity jurisdictions, controlling for enabling technology (smart meter penetration), renewable electricity share, household electricity prices, and GDP per capita. Conditions 1--3 are satisfied uniformly across the sample by the physics of the electrical grid: frequency equilibrium exists in all 30 countries, consumer discretion over consumption timing is universal, and frequency deviation costs are universal. The only variable across countries is C4 --- the degree to which balancing market revenue recycles to demand-side flexibility providers. The prediction is that C4 variation should predict intermediary emergence and that enabling technology should not.

That is what the data shows. Regulatory market openness is the only significant predictor of independent aggregation existence (point-biserial $r = 0.659$, $p \< 0.0001$). Smart meter penetration has zero predictive power ($r = 0.111$, $p = 0.56$). Countries with near-universal smart meter deployment but closed balancing markets --- Spain at 100%, Italy at 100%, Estonia at 98% --- have no independent aggregation. Countries with minimal smart meters but open markets --- Poland at 10%, Germany at 3% --- do. In a Fisher exact test, Maturing markets are 21.6 times more likely to have independent aggregation than non-Maturing markets ($p = 0.0045$). Temporal evidence from 18 countries assessed in both 2017 and 2024 shows market opening preceded intermediary emergence in every improving case, with regulatory changes traceable to exogenous supranational mandates (EU Directive 2019/944, ENTSO-E platform integration) rather than domestic industry lobbying. The test establishes the predictive power of C4 over the technology alternative. It does not uniquely identify the induced symbiosis mechanism --- any model predicting that institutional market access determines firm entry more than technology does would generate the same cross-sectional pattern. The deposit-refund case study below resolves this identification problem.

7.3c Mechanism Case Study: Deposit-Refund Systems

Deposit-refund systems for beverage containers resolve the identification problem of §7.3b by providing a domain where every theoretical variable is directly observable rather than proxied. The companion case study (Buday, 2026d) maps each condition onto the institutional structure. Condition 1 (natural equilibrium): full material circulation is physically defined --- a 100% return rate means all material circulates, and the deviation metric is directly measured by (1 − return rate). Condition 2 (participant discretion): consumers exercise genuine discretion over whether to return containers; return rates below 100% across all countries confirm the choice is real. Condition 3 (genuine systemic cost): litter, landfill, and virgin extraction costs scale with deviation from equilibrium. Condition 4 (loop closure): in a well-designed DRS, the loop is explicitly and completely closed --- unredeemed deposits and material sales fund the collection infrastructure, with no proxy or estimation required.

Norway's 1993 environmental tax on beverage containers implements a cost function that is not merely Pigouvian but explicitly convex in statute. Below 25% return rate: full tax. Between 25% and 95%: tax inversely proportional to return rate, so the marginal cost of each additional unit of deviation depends on aggregate distance from equilibrium. Above 95%: zero tax. This is the cost geometry of Definition 2 written into law. The revenue --- unredeemed deposits plus material commodity value --- funds Infinitum, the not-for-profit

central system operator, covering 95% of operating costs in 2021. The loop is closed by construction. The formal mapping to Definition 3 is exact: the deviation charge is the deposit, the restoration operator Ψ is the collection infrastructure, and the revenue-routing map Φ is the statutory recycling fund mechanism.

The intermediary class predicted by Proposition 3 emerged endogenously. Infinitum's incentive structure is identical to the system health function by construction: higher return rates reduce environmental tax liability for its producer-owners. TOMRA (reverse vending machine manufacturer) earns more as DRS expands and return rate targets rise, aligning its profit function with system health. Collection and recycling firms earn more when more containers are returned. Each intermediary's revenue correlates positively with the return rate, which is the system health metric. This satisfies Definition 5 directly: the geometry makes correct behaviour the only sustainable margin. The cross-country evidence is consistent with the emergence prediction: every European country with a legislated DRS achieves return rates above 85% within a few years of implementation. Countries without DRS achieve 40--55% through kerbside collection despite comparable GDP, technology, and environmental policy. Lithuania's transition from 34% to 92% within two years of DRS implementation --- with the intermediary infrastructure of over 1,000 reverse vending machines emerging in the same window --- demonstrates the rapidity of Proposition 3's convergence prediction when loop closure is achieved. The case study establishes mechanism consistency: in the domain where every theoretical variable is directly observable, the system behaves exactly as the theory predicts.

7.3d Prospective Predictions

Two predictions are registered for evaluation against future data. Prediction 1 (deposit-refund convergence): The EU Packaging and Packaging Waste Regulation, adopted December 2024, requires all member states to implement DRS by 2029. The framework predicts that countries currently without DRS --- France, Italy, Spain, Poland --- will show rapid intermediary emergence and return rate convergence above 85% within one to two years of implementation, consistent with Lithuania (34% to 92% in two years), Slovakia (above 70% in year one), and Ireland (73% in seven months). If this prediction is confirmed across four or more countries, it constitutes strong out-of-sample evidence for the emergence and convergence properties. If falsified, it identifies boundary conditions the theory does not currently account for. Prediction 2 (electricity market intermediary density): As European electricity markets continue opening under the 2019 Electricity Directive, countries transitioning from Early to Maturing market status will show demand-response aggregator

emergence within one to three years of the regulatory change. Countries whose markets remain closed will show no emergence regardless of smart meter penetration or renewable share. Both predictions can be evaluated against publicly available data (DRS implementation outcomes, smartEN/LCP Delta annual assessments) without deployment of a purpose-built mechanism.

7.3e The Evidential Structure

No single piece of evidence is sufficient. The structural audit establishes diagnostic validity: the framework correctly classifies why 60 systems failed, with zero false negatives. The electricity market test establishes the predictive power of C4 over the technology alternative: across 30 countries, the condition the theory identifies as most consequential outperforms the condition conventional policy identifies by a factor of six in explanatory power ($r = 0.659$ vs $r = 0.111$). The deposit-refund case study establishes mechanism consistency: in the domain where every theoretical variable is directly observable, the system behaves exactly as the theory predicts. The prospective predictions establish falsifiability: the PPWR mandate and ongoing electricity market liberalisation provide time-bound, evaluable tests. Together the three pieces address each other's weaknesses. No simpler theory explains all three. The institutional market-access story explains the electricity result but not the mechanism detail of the DRS case. The Pigouvian externality story explains the DRS result but not why C4 --- and not technology --- predicts emergence across countries. The induced symbiosis framework explains both and additionally predicts the audit's failure-mode correspondence. The standard of plausibility appropriate to a formal economic theory with asymptotic claims is met.

7.4 Future Work

Six formal extensions are identified as priorities. First, the multi-agent extension: heterogeneous intermediaries with correlated costs and strategic entry decisions. Proposition 3 uses symmetric intermediaries; under heterogeneous costs, low-cost operators enter first, strengthening rather than weakening the emergence result. The full characterisation of n^* under heterogeneous costs and strategic interaction is the primary formal gap. Second, the finite-N game-theoretic foundation for thin institutional markets such as interbank lending, where the continuum approximation of Appendix B breaks down and the 1/N externality internalization is first-order. Third, the dynamic governance cost extension: treating $C_{open}(t)$ as time-varying makes $d^\dagger(t)$ a moving target and explains why the same level of capture becomes untenable as governance capacity deteriorates. Two structural predictions

are identified as directions for empirical work: designer selection as second-order ESS (§3.5), and Tier 1 mechanisms generating Tier 3 overlays. S* is not specified in advance. It is found by iteration under geometric selection. This paper provides the conditions under which the selection operates, the formal characterisation of what survives it, and the design parameters (α, k, T_career) that determine how fast the selection occurs.

A fourth formal extension concerns agent-type generalisation. The framework assumes agents in the standard economic sense --- utility-maximising, cost-responsive, capable of strategic behaviour. The intermediary class in Linux is not individuals but corporations. As AI systems become economic actors, the intermediary class may include entities whose T_career is not a human lifespan --- possibly near-zero (each inference independent) or possibly very long (a learned policy persists across training runs). The alpha-min argument assumes T_career maps to a human decision-maker's tenure. The framework's predictions shift substantially depending on what kind of agent occupies the intermediary role. The framework's agent-neutrality is one of its strongest features, but it is currently undertheorised. A full treatment is a separate formal contribution.

A fifth concerns the distinction between alignment and selection. The framework produces alignment: agents who behave correctly earn margin. There is a related but distinct mechanism class that produces selection: agents who behave incorrectly are removed. These are not the same thing. Alignment assumes the agent persists and adjusts. Selection assumes the agent is replaced. The Linux case exhibits selection at the patch level --- bad patches are rejected, not reformed --- with alignment operating downstream of that selection. Pulling them apart clarifies what Proposition 3 is doing: emergence is selection-based (entrants appear, non-viable ones exit, viable ones remain); convergence within an emerged mechanism is alignment-based (agents adjust behaviour to track equilibrium). The distinction matters for how the framework's predictions should be evaluated empirically --- selection produces discrete turnover; alignment produces continuous adjustment; the observables differ.

A sixth concerns what the framework structurally cannot predict even if fully developed. These include: whether S* is worth pursuing (a values determination); whether the substrate is worth preserving (a values determination); whether any given domain should have a mechanism at all (a political determination); what happens when multiple frameworks compete for the same substrate (meta-level selection); and whether the framework's own predictions change the behaviour they predict (reflexivity). Naming these limits shifts the framework from a theory that implicitly claims comprehensive coverage to one that explicitly

states its domain. Readers evaluating whether the framework applies to their problem can check against the stated limits rather than discovering them through failed application.

References

Baker, G., Gibbons, R., and Murphy, K. J. (2002). Relational Contracts and the Theory of the Firm. Quarterly Journal of Economics, 117(1), 39--84.

Bergemann, D., and Morris, S. (2005). Robust Mechanism Design. Econometrica, 73(6), 1771--1813.

Hurwicz, L. (1973). The Design of Mechanisms for Resource Allocation. American Economic Review, 63(2), 1--30.

Lasry, J.-M., and Lions, P.-L. (2007). Mean Field Games. Japanese Journal of Mathematics, 2(1), 229--260.

Levin, J. (2003). Relational Incentive Contracts. American Economic Review, 93(3), 835--857.

Ostrom, E. (1990). Governing the Commons: The Evolution of Institutions for Collective Action. Cambridge University Press.

Peltzman, S. (1976). Toward a More General Theory of Regulation. Journal of Law and Economics, 19(2), 211--240.

Rochet, J.-C., and Tirole, J. (2003). Platform Competition in Two-Sided Markets. Journal of the European Economic Association, 1(4), 990--1029.

Stigler, G. J. (1971). The Theory of Economic Regulation. Bell Journal of Economics and Management Science, 2(1), 3--21.

Bergemann, D., and Välimäki, J. (2010). The Dynamic Pivot Mechanism. Econometrica, 78(2), 771--789.

Cox, M., Arnold, G., and Villamayor Tomás, S. (2010). A Review of Design Principles for Community-based Natural Resource Management. Ecology and Society, 15(4), 38.

Bovenberg, A. L., and de Mooij, R. A. (1994). Environmental Levies and Distortionary Taxation. American Economic Review, 84(4), 1085--1089.

d'Aspremont, C., and Gérard-Varet, L. A. (1979). Incentives and Incomplete Information. Journal of Public Economics, 11(1), 25--45.

Goulder, L. H. (1995). Environmental Taxation and the Double Dividend: A Reader\'s Guide. International Tax and Public Finance, 2(2), 157--183.

Green, J., and Laffont, J.-J. (1977). Characterization of Satisfactory Mechanisms for the Revelation of Preferences for Public Goods. Econometrica, 45(2), 427--438.

Maskin, E. (1999). Nash Equilibrium and Welfare Optimality. Review of Economic Studies, 66(1), 23--38.

Maynard Smith, J. (1982). Evolution and the Theory of Games. Cambridge University Press.

Moore, J., and Repullo, R. (1988). Subgame Perfect Implementation. Econometrica, 56(5), 1191--1220.

Myerson, R. B. (1981). Optimal Auction Design. Mathematics of Operations Research, 6(1), 58--73.

Pavan, A., Segal, I., and Toikka, J. (2014). Dynamic Mechanism Design: A Myersonian Approach. Econometrica, 82(2), 601--653.

Schmeidler, D. (1973). Equilibrium Points of Nonatomic Games. Journal of Statistical Physics, 7(4), 295--300.

Appendix A --- Numerical Validation of Stochastic Confinement Bound

Simulation parameters: k = c = 1, $\alpha \in \{1.5, 2.0, 3.0, 3.5\}$, $\sigma \in$ \[0.1, 2.0\]. The theoretical confinement bound \|x_∞\| \~ $(\sigma^2/(2kc))$\^(1/(α+1)) was validated across the full parameter range.

Empirical prefactor C is α-dependent and decreasing: C(1.5) ≈ 0.863, C(2.0) ≈ 0.804, C(3.0) ≈ 0.717, C(3.5) ≈ 0.693, with approximate linear relationship $C(\alpha) \approx -0.086\alpha + 0.983$. Safety margins range from 13.7% at $\alpha = 1.5$ to 30.7% at $\alpha = 3.5$. The bound holds throughout the specified operational range. Full simulation code and figures are available at the repository linked in the paper\'s SSRN record.

Appendix B --- Strategic Foundation of the Aggregate Dynamics

The paper's primary equilibrium concept is evolutionary stability (§3.5). This appendix establishes a supporting result: the aggregate deviation dynamics are derived from individual dominant strategy behaviour in the continuum game, which strengthens the ESS result --- dominant strategy play is sufficient for ESS convergence in the deviation game. Strategic agents with heterogeneous private benefits, each optimising individually under the convex cost geometry, produce the aggregate dynamics the Lyapunov proof operates on. The geometry does not require agents to be cooperative, altruistic, or aware of the mechanism's design. It requires only that each agent responds rationally to the cost signal.

B.1 Setup

Consider a non-atomic continuum of agents indexed by $i \in [0, 1]$ with unit mass. Each agent has a private benefit b_i drawn independently from distribution F on $[b_{min}, b_{max}]$ with $b_{min} \geq 0$. Agent i chooses deviation intensity $a_i \in [0, 1]$, where $a_i = 0$ denotes full equilibrium behaviour and $a_i = 1$ denotes maximum deviation. Aggregate deviation is $x = \int_0^1 a_i \, di$. Because agents are non-atomic --- each individual has measure zero in the continuum --- no single agent's choice a_i alters the aggregate state x. Each agent therefore takes $\tau(x)$ as a genuinely exogenous variable, not as a Nash conjecture approximation. This connects to the Mean Field Game literature (Lasry and Lions 2007) cited in §1.1, where the continuum structure is standard. Under the closed convex loop, the cost rate imposed on deviation is $\tau(x) = c \cdot |x|^\alpha$ for $\alpha > 1$, increasing and convex in x. Agent i's net payoff decomposes as $\pi_i(a_i, x) = V_i(a_i) - C_i(a_i, x)$, where $V_i(a_i) = b_i \cdot a_i$ is the private benefit depending only on agent i's own action --- not on other agents' actions and not on aggregate x --- and $C_i(a_i, x) = \tau(x) \cdot a_i$ is the cost imposed by the mechanism. Strategic interaction cannot enter through the benefit side because V_i is separable in a_i. The dominant strategy derivation below is exact in the continuum formulation.

B.2 Dominant Strategy Characterisation

The cost component $C_i(a_i, x) = \tau(x) \cdot a_i = c \cdot |x|^\alpha \cdot a_i$ is convex in a_i for all $x \geq 0$ simultaneously --- not merely at equilibrium, but across every aggregate state the system could occupy. Since $\tau(x) \geq 0$ for all $x \geq 0$, the marginal cost of deviation is $\tau(x)$ at every out-of-equilibrium state, every transition, and at equilibrium. Because C_i is linear (hence convex) in a_i for all x, and V_i is linear in a_i and independent of x, the net payoff π_i is linear in a_i for all x. Agent i's best response is a corner solution for every x: $a_i^* = 1$ if $b_i > \tau(x)$, $a_i^* = 0$ if $b_i < \tau(x)$. This is a dominant strategy in the continuum sense: the decision rule $a_i^*(x) = 1\{b_i > \tau(x)\}$ is ex post optimal for every realised aggregate state x, regardless of what other agents do, before and after equilibrium, at every aggregate state. Formally, this corresponds to a dominant strategy equilibrium in the sense of the price-taking equilibrium standard in non-atomic games (Schmeidler 1973, Wardrop 1952): no agent can profitably deviate unilaterally for any realisation of the aggregate state. Strategic interaction cannot alter this conclusion because the dominance condition holds globally in x, not merely at a fixed point. For finite N, each agent internalises 1/N of the cost externality their deviation imposes on the aggregate. The maximum gain from strategic deviation above the threshold $\tau(x)$ is bounded by $c \cdot \alpha \cdot |x|^{(\alpha-1)} \cdot (1/N)$, which is $O(1/N)$ and negligible for large participant populations. For thin markets with small N --- such as interbank lending where N

≈ O(10--50) --- this bound is non-negligible and the dominant strategy result is approximate, as noted in §5.2.

B.3 Aggregate Dynamics Derived

Given dominant threshold strategies, the equilibrium fraction deviating is $1 - F(\tau(x))$. Nash equilibrium aggregate deviation satisfies $x = 1 - F(c \cdot \|x\|^{\alpha})$. Define $g(x) = 1 - F(c \cdot \|x\|^{\alpha}) - x$. At $x = 0$: $g(0) = 1 - F(0) = 1 > 0$ since $b_{min} \geq 0$ implies $F(0) = 0$. At $x = 1$: $g(1) = -F(c) \leq 0$. By continuity a fixed point $x^* \in [0,1]$ exists. Since $x \to 1 - F(\tau(x))$ is strictly decreasing in x (F strictly increasing, τ strictly increasing), x^* is unique. Under Condition 4, collected costs return to the restoration operator, generating continuous adjustment dynamics. The static fixed-point analysis above establishes what aggregate state individual strategies produce at any moment. The dynamic ODE arises from the restoration operator Ψ, which under Condition 4 receives collected cost revenue and deploys it to reduce $\|x\|$ at rate k. The rate of change of aggregate deviation is therefore determined by how fast the restoration operator acts on the deviation level that individual strategies produce. For the continuum system in continuous time: $dx/dt = -k \cdot c \cdot \|x\|^{\alpha} \cdot \mathrm{sign}(x)$. This is the aggregate ODE of §3.1a, now derived from individual dominant strategy optimisation (which determines the aggregate state) combined with the restoration dynamics of Condition 4 (which determines how that state evolves over time) rather than assumed. The Lyapunov proof then establishes asymptotic stability. Non-cooperative agents with heterogeneous types produce cooperative aggregate convergence as a consequence of individual rationality. The geometry does not ask agents to be good. It makes being good the dominant strategy.

B.4 Scope

This appendix establishes, within a non-atomic continuum game with heterogeneous private types, that the aggregate dynamics of §3.1a are derived from dominant strategy behaviour for any distribution F satisfying $b_{min} \geq 0$. It does not model strategic interaction between intermediaries in the entry game of Proposition 3 --- that extension remains future work. It does not address repeated game dynamics or correlated private information. Regarding collusion: a coalition of intermediaries maintaining above-competitive spreads through non-economic entry suppression satisfies Definition 7 (the captured loop) at the intermediary level. The identification criterion applies --- the coalition opposes recalibration of the spread target --- and the captured loop analysis of §4.2 encompasses this as a named failure mode. Coalitional deviation is therefore not an unaddressed gap in the ESS result; it is a form of capture whose diagnosis and identification criterion are already in the paper. Within its

stated scope the derivation is complete, and the main text's claim that the geometry produces dominant strategy behaviour is now formally established rather than stated by analogy. The continuum formulation is essential: in a finite game with N agents, agent i's choice a_i shifts x by 1/N, making $\tau(x)$ endogenous to i's action and breaking the dominant strategy argument. In the non-atomic continuum, each agent has measure zero and x is genuinely exogenous, making the dominant strategy result exact rather than approximate. A separate note on Jacobian linearisation: when $\alpha > 1$, the spring constant $\Phi_x = knc\alpha|x|^{(\alpha-1)}$ evaluates to zero at the origin, making Jacobian linearisation at x = 0 invalid. This paper avoids Jacobian linearisation entirely, using Lyapunov methods directly on the nonlinear system. The stochastic confinement bound of §3.1b is the correct characterisation of near-origin behaviour: the system is confined to a neighbourhood of radius proportional to σ, not driven to exactly zero.